

Chabauty–Kim Methods over Number Fields:

$$\mathbb{P}^1 \setminus \{0, 1, \infty\}$$

Xiang Li

(joint work with Martin Lüdtkke)

University of Edinburgh

Motivic Chabuty–Kim Methods

Let $X = \mathbb{P}^1 \setminus \{0, 1, \infty\}$, K be a number field, p split completely in K , Π be a quotient of $\pi_1^{\text{mot}}(X, b)$. Write $\pi_1(\text{MT}(\mathcal{O}_{K,S}, \mathbb{Q})) = U_S^{\text{MT}} \rtimes \mathbb{G}_m$.

$$\begin{array}{ccc}
 X(\mathcal{O}_{K,S}) & \hookrightarrow & X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p) = \prod_{p|p} X(\mathcal{O}_p) \\
 \downarrow j_S^{\text{mot}} & & \downarrow j_p^{\text{dR}} = \prod_p j_p^{\text{dR}} \\
 \text{Sel}_{S,\Pi}^{\text{mot}}(X)_{\mathbb{Q}_p} & \xrightarrow{\prod_p \text{ev}_{\eta_p}} & \prod_{p|p} \text{Res}_{K_p/\mathbb{Q}_p} \Pi_{K_p}^{\text{dR}}.
 \end{array}$$

Polylogarithmic Quotient $\text{Lie}(\Pi_{\text{PL}}) \simeq \mathbb{Q}(1) \rtimes \prod_{n=1}^{\infty} \mathbb{Q}(n)$.

$$\text{Sel}_{S,\text{PL},N}^{\text{mot}}(X) \simeq \text{Hom}_{\text{gr}}(\text{Lie}(U_S^{\text{MT}}), \text{Lie}(\Pi_{\text{PL},N}^{\omega})).$$

$$L_n(z) = \sum_{k=0}^{n-1} \frac{B_k}{k!} \log(z)^k \text{Li}_{n-k}(z)$$

CK Locus and Kim's conjecture

Definition (Chabauty–Kim locus)

$$X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi} := j_p^{-1}(\mathrm{loc}_p(\mathrm{Sel}_{S,\Pi}^{\mathrm{mot}}(X)_{\mathbb{Q}_p})) \subseteq X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p),$$
$$X(\mathcal{O}_{\mathfrak{p}})_{S,\Pi} := \mathrm{pr}_{\mathfrak{p}}(X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi}) \subseteq X(\mathcal{O}_{\mathfrak{p}}).$$

Another candidate for single prime locus, which is too large:

$$X(\mathcal{O}_{\mathfrak{p}})'_{S,\Pi} := j_p^{-1}(\mathrm{loc}_{\mathfrak{p}}(\mathrm{Sel}_{S,\Pi}^{\mathrm{mot}}(X)_{\mathbb{Q}_p})) \subseteq X(\mathcal{O}_{\mathfrak{p}}).$$

Conjecture (Kim's Conjecture)

$$X(\mathcal{O}_K \otimes_{\mathbb{Z}} \mathbb{Z}_p)_{S,\Pi} = X(\mathcal{O}_{K,S}).$$
$$X(\mathcal{O}_{\mathfrak{p}})_{S,\Pi} = X(\mathcal{O}_{K,S}).$$

Roots of Unity

Theorem

K : number field containing an imaginary quadratic field;
 S : containing all primes of K lying over some rational prime l ;
 $\mathfrak{p}$: dividing $p \neq l$ splitting completely in K .
Assume the p -adic period conjecture.
 $\Rightarrow X(\mathcal{O}_{\mathfrak{p}})_{S, \text{PL}, \infty}$ contains all roots of unity $\neq 1$ in $\mathbb{Z}_p$.

Corollary

In the situation of the above theorem, Kim's Conjecture fails for the polylogarithmic quotient when p is large.

Example: $K = \mathbb{Q}(\zeta_8)$, $S = \{(1 - \zeta_8)\}$

Equations (up to depth 2):

$$\begin{aligned}\log(z_1) &= \log(z_2), & \log(z_3) &= \log(z_4), \\ \mathrm{Li}_1(z_1) &= \mathrm{Li}_1(z_2), & \mathrm{Li}_1(z_3) &= \mathrm{Li}_1(z_4), \\ L_2(z_1) + L_2(z_2) + L_2(z_3) + L_2(z_4) &= 0, \\ L_2(z_1) + L_2(z_2) &= c(-\log(z_3)\mathrm{Li}_1(z_1) + \log(z_1)\mathrm{Li}_1(z_3)).\end{aligned}$$

Results:

- locus = actual solutions + roots of unity for $p < 200$.
- For $p < 200$ where $\zeta_6 \notin \mathbb{Z}_p$: **full quotient** Kim's conjecture holds.

Quadratic Field: Verify Kim's Conjecture

K	S	p	polylog	full
$\mathbb{Q}(\zeta_3)$	$\emptyset$	$p \equiv 1 \pmod{3}$	holds	holds
$\text{imag quad} \neq \mathbb{Q}(\zeta_3)$	$\emptyset$	$p \not\equiv 1 \pmod{3}$	holds	holds
$\text{imag quad} \neq \mathbb{Q}(\zeta_3)$	$\emptyset$	$p \equiv 1 \pmod{3}$	fails	
$\mathbb{Q}(\sqrt{-2})$	$\{(\sqrt{-2})\}$	$p = 3$	holds	holds
$\mathbb{Q}(\sqrt{-2})$	$\{(\sqrt{-2})\}$	$p > 3$	fails	
$\mathbb{Q}(i)$	$\{(1-i)\}$	$p = 5$	holds	holds
$\mathbb{Q}(i)$	$\{(1-i)\}$	$p > 5$	fails	
$\mathbb{Q}(\zeta_3)$	$\{(1-\zeta_3)\}$	$p = 7$	holds	holds
$\mathbb{Q}(\zeta_3)$	$\{(1-\zeta_3)\}$	$7 < p < 2000$	fails	holds
$\mathbb{Q}(\sqrt{5})$	$\emptyset$	$p < 2000$	holds	holds
$\text{real quad} \neq \mathbb{Q}(\sqrt{5})$	$\emptyset$	$p \not\equiv \pm 1 \pmod{5}$ and $p < 2000$	holds	holds
$\mathbb{Q}(\sqrt{2})$	$\emptyset$	$p < 2000$	holds	holds
$\mathbb{Q}(\sqrt{2})$	$\{(\sqrt{2})\}$	$p < 100$	holds	holds

Refined Locus and S -Selmer Section Conjecture

Theorem

K : imaginary quadratic field

$S = \{\mathfrak{l}\}$ which is not $\text{Gal}(K/\mathbb{Q})$ -stable.

p : splitting completely in K , not divisible by $\mathfrak{l}$

$$\Rightarrow X(\mathcal{O}_K \otimes \mathbb{Z}_p)_{S, \text{PL}, \infty}^{(1)} = \{(\zeta_6, \zeta_6^{-1}), (\zeta_6^{-1}, \zeta_6)\} \text{ (if } \zeta_6 \in \mathbb{Z}_p).$$

Conjecture (S -Selmer Section Conjecture)

$$X(\mathcal{O}_{K,S}) = \text{Sec}(X/\mathcal{O}_{K,S})^{\text{Sel}}.$$

Theorem

The S -Selmer Section Conjecture holds for K an imaginary quadratic field, S which is empty or contains precisely one prime which is not stable under the $\text{Gal}(K/\mathbb{Q})$ -action, and the curve $X = \mathbb{P}^1 \setminus \{0, 1, \infty\}$.